Project 2: Power BI Data Modeller

1. Schema Type: Star Schema

In this project, I used a star schema design where Sales_Fact acts as the central fact table. All the dimension tables are directly connected to it, which keeps the model simple and improves performance in Power BI.

I also included Returns_Fact as a second fact table instead of treating it as a dimension. It is connected to Sales_Fact through SalesID and shares the Date_Dim table. There are two relationships with Date_Dim:

one active (using DateKey)

one inactive (using ReturnDateKey)

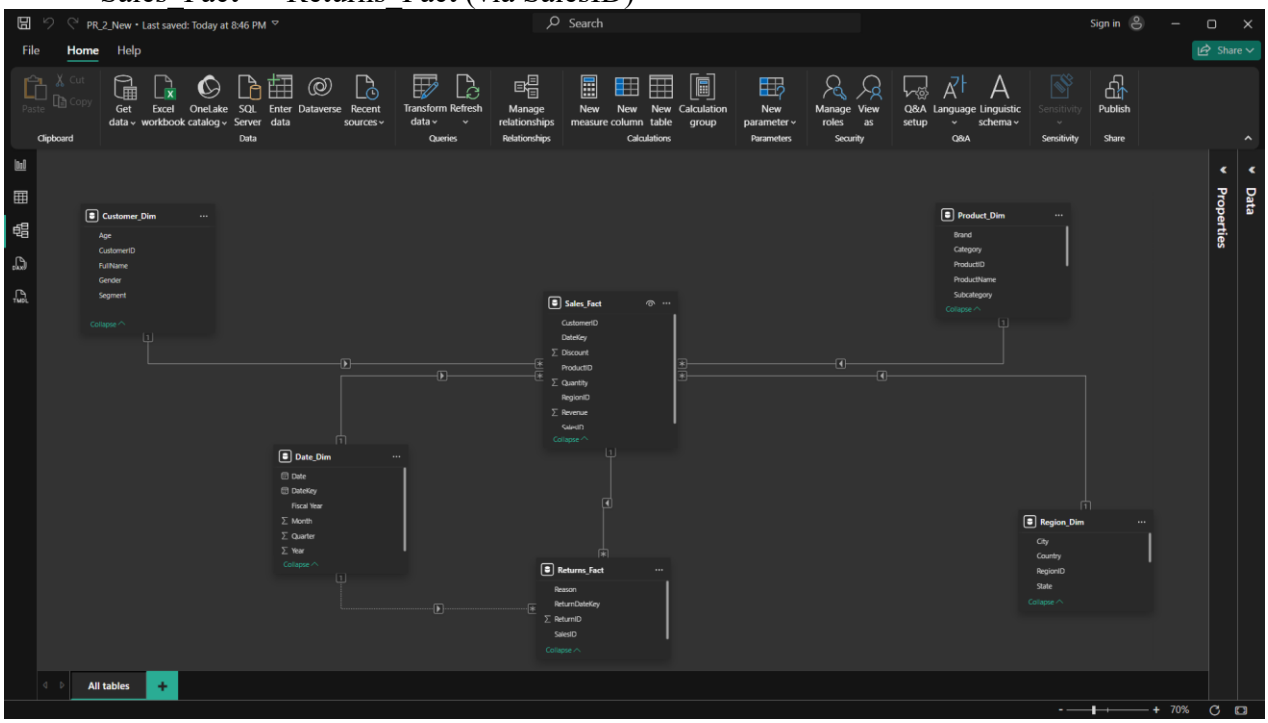
2. Relationships and Filter Flow

2a. Active Relationships

All relationships between dimension tables and fact tables are set to one-to-many (1:M) with single-direction filtering (Dimension → Fact). This helps avoid confusion and ensures filters behave as expected.

The main relationships are:

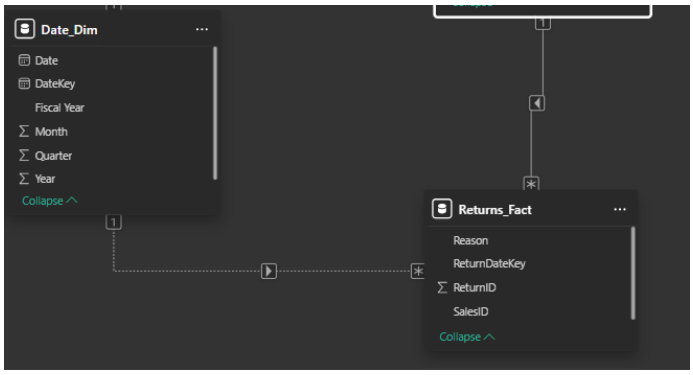
Customer_Dim → Sales_Fact (via CustomerID)
Product_Dim → Sales_Fact (via ProductID)
Region_Dim → Sales_Fact (via RegionID)
Date_Dim → Sales_Fact (via DateKey)
Sales_Fact → Returns_Fact (via SalesID)



There is also a relationship between Date_Dim and Returns_Fact using ReturnDateKey, but this is kept inactive.

2b. Inactive Relationship (Returns_Fact – ReturnDateKey)

The relationship between Date_Dim[DateKey] and Returns_Fact[ReturnDateKey] is set as inactive. This is done to avoid ambiguity, since Power BI allows only one active relationship path between tables.



2c. Bidirectional Filters

I have not used bidirectional filtering in this model. All relationships use single-direction filtering to avoid circular dependencies and unexpected results.

Bidirectional filtering is only useful in special cases like many-to-many relationships, which are not present in this model.

3. Data Categories and Hierarchies

To make reporting easier, I created hierarchies and assigned proper data categories:

Region_Dim

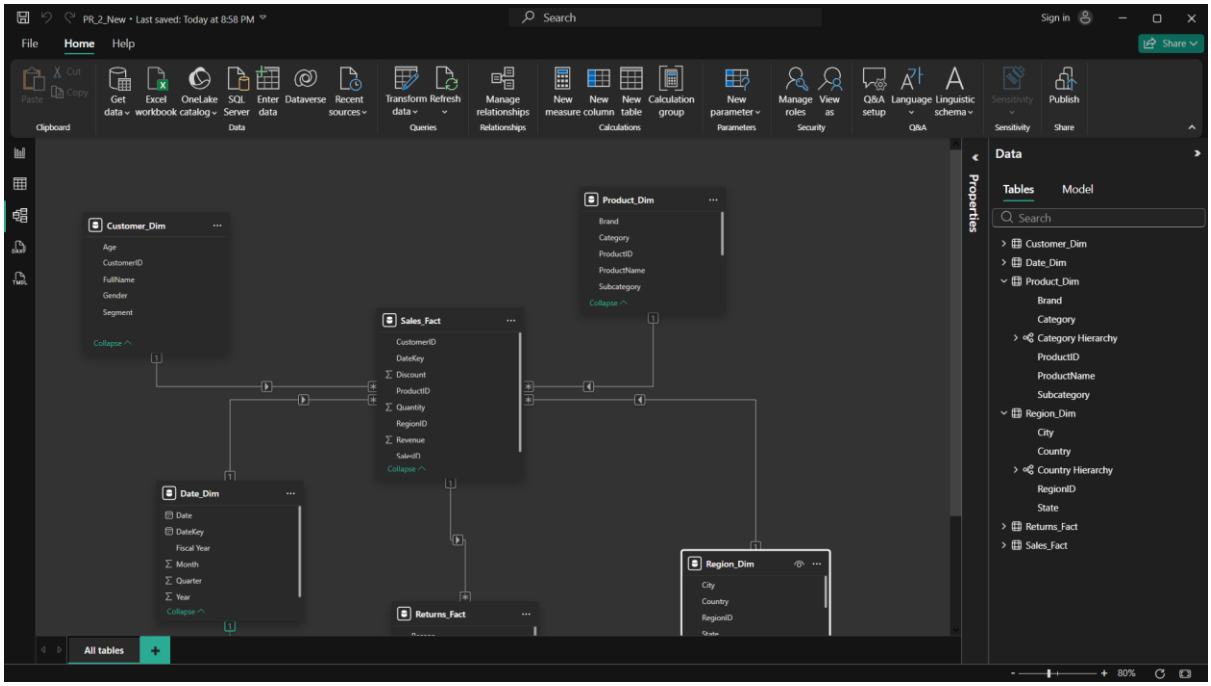
Country (Country category)
State (State/Province)
City (City)
Hierarchy: Country → State → City

Date_Dim

Hierarchy: Year → Quarter → Month → Date

Product_Dim

Hierarchy: Category → Subcategory → Product Name



These hierarchies help with drill-down analysis in visuals.

5. Verification Using Matrix Visual

To check if the model was working correctly, I created a Matrix visual and tested different scenarios:

Sales by Product Category and Region → confirmed data is showing correctly across all categories and regions

Returns by Fiscal Year → verified that USERELATIONSHIP() works for return date analysis

Revenue by Customer Segment → confirmed filters from Customer_Dim are applied properly

Category	Germany	India	USA	Total
Clothing	26521	27725	38015	**92261**
Electronics	26489	31692	44712	**102893**
Furniture	23990	26347	41435	**91772**
Total	**77000**	**85764**	**124162**	**286926**

Segment	Sum of Revenue
Gold	95597
Platinum	93893
Silver	97436
Total	**286926**

Reason	Count of ReturnID
Wrong Item	31
Damaged	34
Not Needed	35
Total	**100**

Country
- Germany
- India
- USA

State
All

Year
- 2022
- 2023

Month

1	3	5	7	9	11
2	4	6	8	10	12

Add Slicer and Apply Slicer

Category	India	Total
Clothing	1039	**1039**
Electronics	1680	**1680**
Furniture	790	**790**
Total	**3509**	**3509**

Segment	Sum of Revenue
Gold	984
Platinum	1022
Silver	1503
Total	**3509**

Reason	Count of ReturnID
Damaged	1
Not Needed	1
Wrong Item	2
Total	**4**

Country

- [x] India
- [ ] USA

State

TX

Year

- [x] 2022
- [ ] 2023

Month

1	3	5	7	9	11
2	4	6	8	10	12